

EE-312L Signals and Systems Lab
Laboratory Session 06
**To Understand and Demonstrate the
Linearity Property of a System Using
MATLAB**

October 23, 2024

Outlines:

1. Objective
2. Linearity Property of Continuous-Time Systems and Discrete-Time Systems
3. Use MATLAB to verify Linearity Property

1 Objective

The objective of this lab session is to understand and demonstrate the Linearity property of continuous-time systems and discrete-time systems using MATLAB.

2 Linearity Property of Continuous-Time Systems and Discrete-Time Systems

- A linear system, in continuous-time or discrete-time, is a system that possesses the important property of superposition.
- If an input consists of the weighted sum of several signals, then the output is the superposition – that is, the weighted sum – of the responses of the system to each of those signals.

2.1 Linearity Property of Continuous-Time Systems

- Let $y_1(t)$ be the response of a continuous-time system to an input $x_1(t)$
- Let $y_2(t)$ be the response of a continuous-time system to an input $x_2(t)$
- If we apply the input $ax_1(t) + bx_2(t)$ to the system (linear), the response will be $ay_1(t) + by_2(t)$. Where a and b are constants as shown in Figure 1.

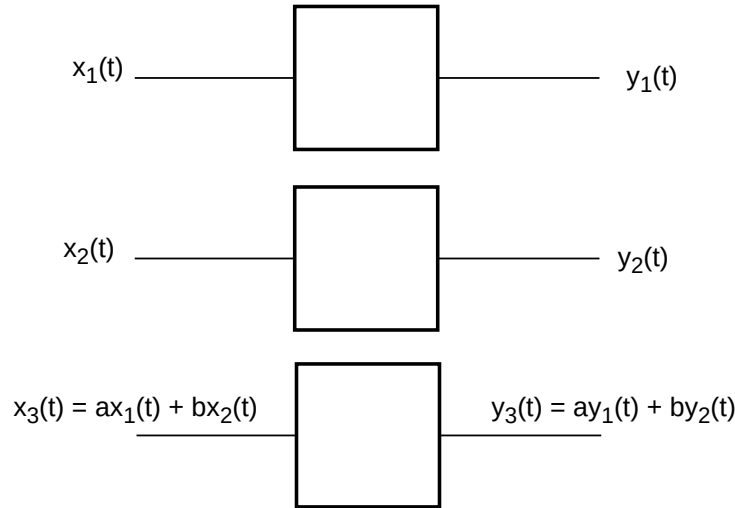


Figure 1: Linearity in continuous-time systems

2.2 Linearity Property of Discrete-Time Systems

- Let $y_1[n]$ be the response of a discrete-time system to an input $x_1[n]$
- Let $y_2[n]$ be the response of a discrete-time system to an input $x_2[n]$
- If we apply the input $ax_1[n] + bx_2[n]$ to the system (linear), the response will be $ay_1[n] + by_2[n]$. Where a and b are constants Figure 2.

3 Use of MATLAB to Verify Linearity Properties

Example 6.1:

Let $x_1(t) = u(t) - u(t - 1)$ and $x_2(t) = u(t) - u(t - 2)$ be input signals to the systems described by the input-output relationships as follows

$$y(t) = 2x(t) \quad (1)$$

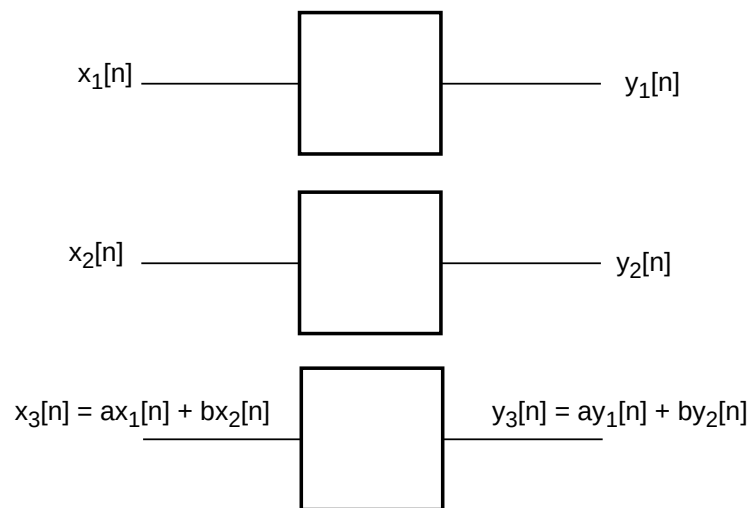


Figure 2: Linearity in discrete-time systems

Determine if the linearity property holds for this system. To examine if the system is linear, we choose the scalars $a_1 = 2$ and $a_2 = 3$.

```

1 clear all; close all; clc;
2 t=-3:0.01:3;
3 x1 = heaviside(t)-heaviside(t-1);
4 x2 = heaviside(t)-heaviside(t-2);
5
6 a = 2;
7 b = 3;
8
9 x3 = a.*x1 + b.*x2;
10
11 %% for y(t) = 2*x(t)
12
13
14
15 y1 = 2*x1;
16 y2 = 2*x2;
17 y3 = 2*x3;
18
19
20 figure(1)
21 subplot(3,1,1);
22 plot(t,x1, 'r', 'linewidth', 2.5);
23 xlabel('t');
24 ylabel('x_{1}(t)');
```

```
25 ylim([-0.2 5.2]);
26 xlim([-3 3]);
27 legend('x_{1}(t)');
28 grid minor on;
29
30 subplot(3,1,2);
31 plot(t,x2, 'linewidth', 2.5);
32 xlabel('t');
33 ylabel('x_{2}(t)');
34 ylim([-0.2 5.2]);
35 xlim([-3 3]);
36 legend('x2(t)');
37 grid minor on;
38
39 subplot(3,1,3);
40 plot(t,x3, 'linewidth', 2.5);
41 xlabel('t');
42 ylabel('x_{3}(t)');
43 ylim([-0.2 5.2]);
44 xlim([-3 3]);
45 legend('x3(t)');
46 grid minor on;
47
48 saveas(gcf, 'lab06plots/plot01.pdf')
49
50
51 figure(2)
52 subplot(2,1,1);
53 plot(t,y1, 'r', 'linewidth', 2.5);
54 xlabel('t');
55 ylabel('y_{1}(t)');
56 ylim([-0.2 5.2]);
57 xlim([-3 3]);
58 legend('y_{1}(t)');
59 grid minor on;
60
61 subplot(2,1,2);
62 plot(t,y2, 'linewidth', 2.5);
63 xlabel('t');
64 ylabel('y_{2}(t)');
65 ylim([-0.2 5.2]);
66 xlim([-3 3]);
67 legend('y_{2}(t)');
68 grid minor on;
69
```

```
70
71 saveas(gcf, 'lab06plots/plot02.pdf')
72
73 figure(3)
74 subplot(2,1,1);
75 plot(t,y3, 'r', 'linewidth', 2.5);
76 xlabel('t');
77 ylabel('y_{3}(t) = 2y_{1}(t) + 3y_{2}(t)');
78 ylim([-0.2 10.2]);
79 xlim([-3 3]);
80 legend('y3(t) = ');
81 grid minor on;
82
83 subplot(2,1,2);
84 plot(t,y3, 'linewidth', 2.5);
85 xlabel('t');
86 ylabel('y_{3}(t) = 2(x_{3}(t))');
87 ylim([-0.2 10.2]);
88 xlim([-3 3]);
89 legend('y_{3}(t) = 2x_{3}(t)');
90 grid minor on;
91
92
93 saveas(gcf, 'lab06plots/plot03.pdf')
```

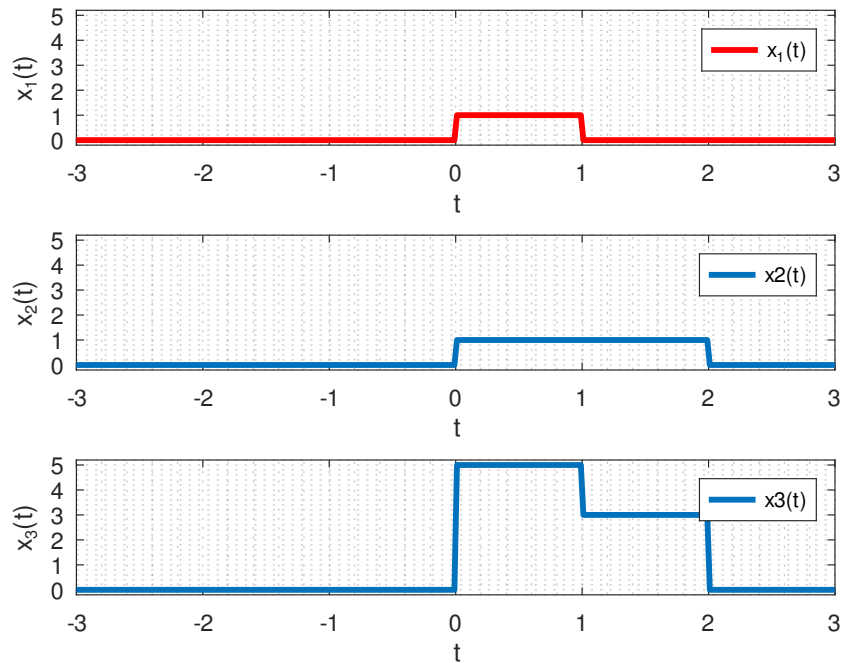
Example 6.2:

Let $x_1(t) = u(t) - u(t - 1)$ and $x_2(t) = u(t) - u(t - 2)$ be input signals to the systems described by the input-output relationships as follows

$$y(t) = x^2(t) \quad (2)$$

2 Determine if the linearity property holds for this system. To examine if the system is linear, we choose the scalars $a_1 = 2$ and $a_2 = 3$.

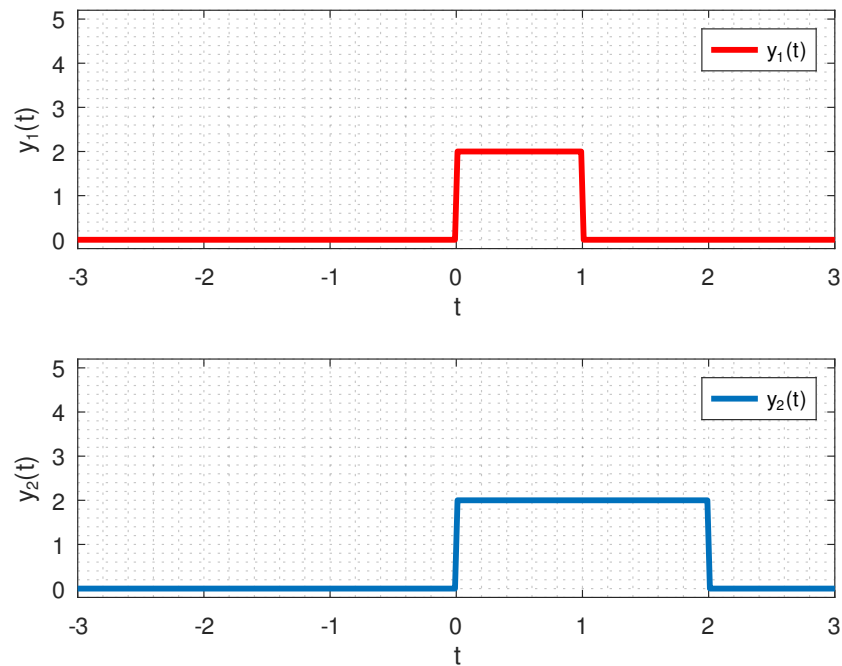
```
1
2 clear all; close all; clc;
3 t=-3:0.01:3;
4 x1 = heaviside(t)-heaviside(t-1);
5 x2 = heaviside(t)-heaviside(t-2);
6
7 a = 2;
8 b = 3;
9
10 x3 = a.*x1 + b.*x2;
```

Figure 3: Input signals $x_1(t)$, $x_2(t)$, and $x_3(t)$

```

11
12 %% for  $y(t) = x(t)^2$ 
13
14
15
16 y1 = x1.^2;
17 y2 = x2.^2;
18 y3 = x3.^2;
19
20
21
22 figure(1)
23 subplot(2,1,1);
24 plot(t,y1, 'r', 'linewidth', 2.5);
25 xlabel('t');
26 ylabel('y_1(t)');
27 ylim([-0.2 5.2]);
28 xlim([-3 3]);
29 legend('y_{1}(t)');
30 grid minor on;
31
32 subplot(2,1,2);
33 plot(t,y2, 'linewidth', 2.5);
34 xlabel('t');

```

Figure 4: Output signals $y_1(t)$ and $y_2(t)$

```

35 ylabel('y_2(t)');
36 ylim([-0.2 5.2]);
37 xlim([-3 3]);
38 legend('y_{2}(t)');
39 grid minor on;
40
41
42 saveas(gcf, 'lab06plots/plot04.pdf')
43
44 figure(2)
45 plot(2,1,1);
46 plot(t,2*y1+3*y2, 'b', 'linewidth', 2.5, t,y3,'r', 'linewidth',
47      2.5);
48 xlabel('t');
49 ylabel('y_{3}(t)');
50 ylim([-0.2 25.4]);
51 xlim([-3 3]);
52 legend('y_{3}(t) = 2y_{1}(t) + 3y_{2}(t)', 'y_{3}(t) = x_
53        {3}^{2}(t)');
54 grid minor on;
55 saveas(gcf, 'lab06plots/plot05.pdf')

```

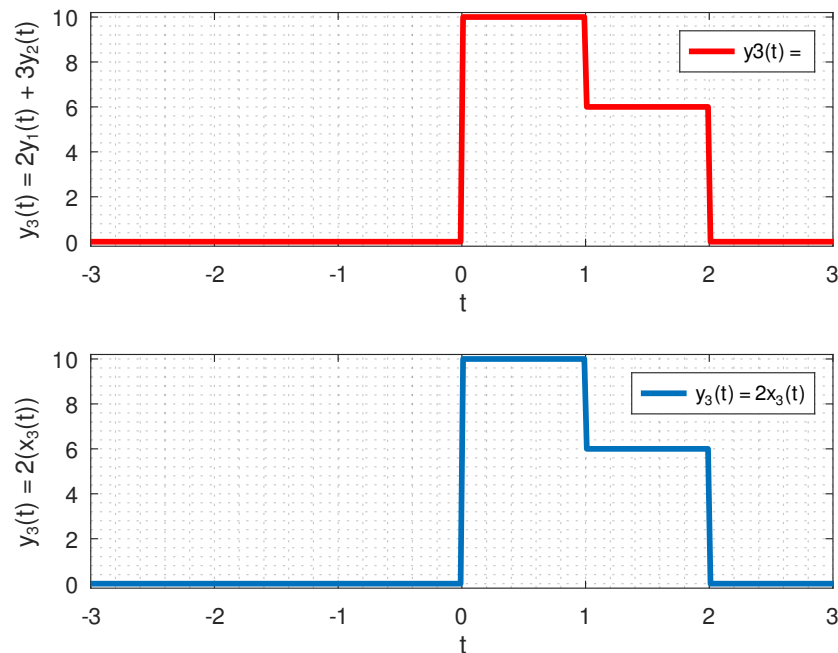
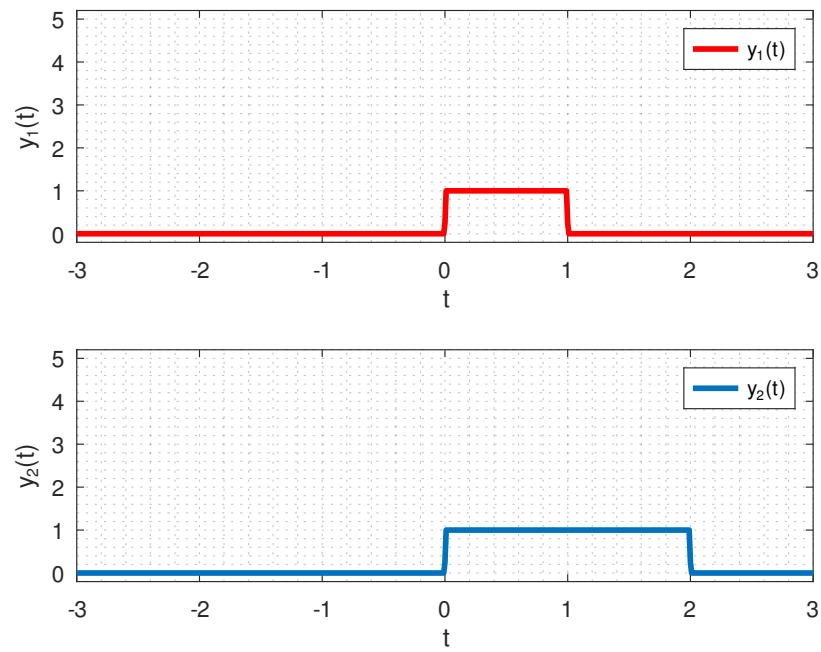
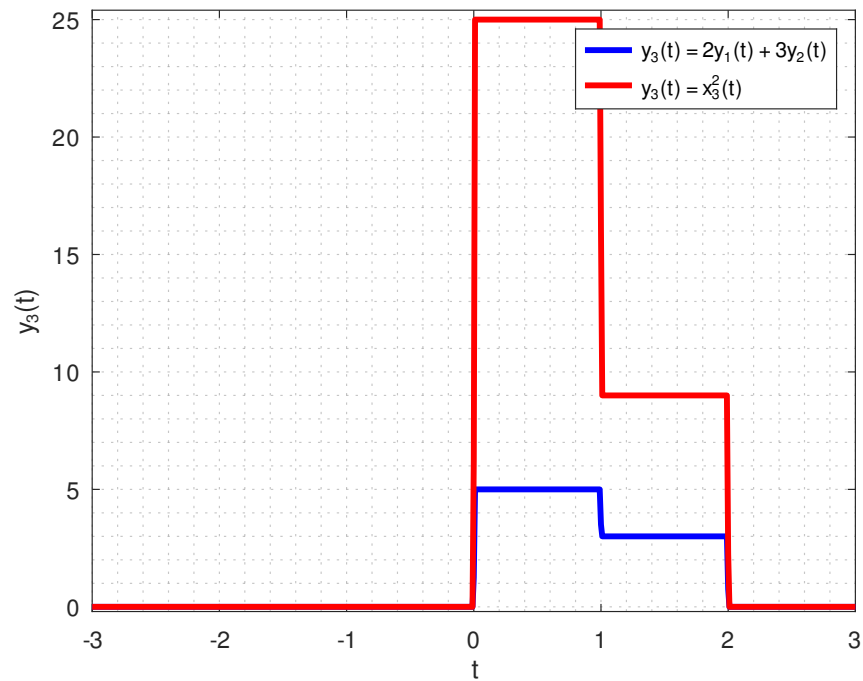


Figure 5: The results of these two graphs agree which means that $y(t) = 2x(t)$ is a linear systems

4 Lab 06: Lab Tasks

1. Use MATLAB to determine whether the system with input-output relationship $y(t) = x(t) + x(t - 1)$ is linear.
2. Use MATLAB to determine if the system with input-output relationship $y(t) = 3x(t) + 2\cos(\pi t/3)$ is linear.
3. Use MATLAB to determine if the system with input-output relationship $y[n] = x^2[n]$ is linear.

Figure 6: Output signals $y_1(t)$ and $y_2(t)$ Figure 7: The results of these two graphs do not agree which means that $y(t) = x^2(t)$ is not a linear systems